

GiBleed Clinical Audit — Unified Reference

A single comprehensive companion to the master record. Deep, first-principles ELI5 treatments of the topics worked through in conversation, integrated with faithful coverage of every remaining finding.

Provenance and precedence. Every number here is taken from the master record, not re-derived. Where this document conflicts with the master record on any figure, **the master record wins**. The deep-dive sections (M1, M2, B1, B3, B4) reflect extended discussion; the remaining sections (M3–M7, B2, B5, §D–§H) are condensed faithfully from the master record with ELI5 layers added.

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0 • The work in one page

You set out to learn OMOP. You ended up recovering an **undocumented module** from a synthetic-data generator using nothing but SQL and clinical knowledge.

GiBleed — the default teaching database for Eunomia, where nearly everyone first meets OHDSI/OMOP — **passes every automated data-quality check and describes a world that cannot exist**. Eight conditions occur exactly once per person and never again. Seven of them, read together, are the **differential diagnosis of GI bleeding**. The eighth is osteoarthritis, which 100% of the population has. All eight stop by age 47 in a database that records other conditions to 109. Nobody dies.

None of it violates a range check. Every value is well-formed, in-range, type-correct, conformant.

The thesis. The mathematics tells you what shape a disease should leave in data. The

medicine tells you when the shape is wrong.

🧑 ELI5 — the whole project A flight simulator is *perfect* for practising landings and *useless* for studying what turbulence does to a body. Neither statement criticises the simulator. The only problem is that **nobody wrote "not for turbulence research" on the box**. That's GiBleed. This audit is the missing label.

A • Orientation

A.1 The three-domain lens

Data quality is usually treated as one question. It is three, and only the first has tooling.

Domain	The question it asks	GiBleed's answer
Computer science	Is the value well-formed?	✅ passes everything
Medicine	Is this what the disease <i>does</i> ?	❌
Biomathematics	Is this a <i>shape</i> a disease can leave?	❌

Automated data-quality tooling — Kahn's conformance/completeness/plausibility framework, the OHDSI Data Quality Dashboard — lives **entirely inside the first circle**.

Every finding in this audit sits in the **Medicine** $\cap$ **Biomathematics** overlap, where no range check has ever looked. The centre of the Venn is *fitness for purpose*, which requires all three — and the OMOP CDM has no field for it.

🧑 ELI5 — why this is a staffing argument, not a personal boast A CS-trained data-quality engineer passes this dataset — *correctly*, by every standard in their toolkit. A clinician without modelling sees the ages are wrong but not that the *shape* is wrong. A modeller without medicine sees `recs_per_person = 1.00` and has no prior that says it should be 6. **The findings live only in the overlap. A team needs all three. That is the whole finding.**

A.2 The two instruments

Everything was recovered with two columns most people treat as bookkeeping:

- `recs_per_person` — how many records each affected person has for a condition. Read correctly, this is a *measurement of the disease's state graph* (§C1).
- **the age envelope** — the min-to-max age at diagnosis. Read correctly, this is a *measurement of the hazard function* (§C3).

Measured **separately**, these two rulers select the *same eight conditions* (§C5). That agreement is the backbone of the whole audit.

A.3 The distinction that must never blur

Three different quantities are constantly confused. Keeping them apart is what makes the findings correct rather than merely striking.

quantity	formula	plain meaning
prevalence	$n_people / 2694$	how many people have it
records per person	$n_records / n_people$	how many times each got it
incidence	new cases per unit time	the rate of new cases (the "tap", §C3)

🧑‍🎓 **ELI5** "Fifteen people came to the party, each once" and "everyone in town came" are different sentences. $recs_per_person = 1.00$ is the **first**. $prevalence = 100\%$ is the **second**. Only osteoarthritis is at 100% prevalence. Ulcerative colitis is $413/2,694 = 15.3\%$ — and all 413 of them got it exactly once.

B • The medical lens

Ordered most-consequential first. Each finding carries the downstream analytic damage it invites.

B — M1 • There is no confounding by indication

The single most serious finding in the audit.

355 of 479 GI bleeders are on celecoxib. Drug precedes bleed 355 times out of 355 — zero reversals, zero same-day ties.

M1.1 The problem the whole field exists to solve

The question is: *does celecoxib cause GI bleeding?* The naive approach — count bleeds among users vs non-users, divide — fails because of **why people got the drug**. A clinician reaches for celecoxib (a COX-2 selective) *because* the patient is already bleed-prone. So the treated group was going to bleed more anyway. The drug got given **to the high-risk**.

This is **confounding by indication**, and it is the entire reason NSAID→GI-bleed is a hard question rather than a division problem. If celecoxib were pure sugar, you would *still* see more bleeds among users, because the fragile people are the ones holding it.

🧑‍🎓 **ELI5 — umbrellas at the beach** The data says umbrella-carriers get rained on more. So umbrellas cause rain! Really, people carry umbrellas **because** rain is likely. Doctors give celecoxib **because** the patient is already bleed-prone. GiBleed deleted the "because" — which makes the puzzle easy, and makes it not the puzzle.

REAL WORLD

GIBBLE

GI risk



Celecoxib -> GI bleed

Risk causes both. That is
the problem the methods solve.

GI risk



Celecoxib -> GI bleed

The risk -> drug arrow
was never drawn.

M1.2 The five methods, and what each actually does

Every method in the finding is an answer to the same question: *how do I get an honest estimate when the treated and untreated groups were never comparable?*

- **Propensity score.** A model predicting, from a person's characteristics, how likely they were to *receive* the drug. That probability (call it e) is the score. Compare treated and untreated people with the **same score** — within that slice, assignment is near a coin flip.
- **IPTW — inverse probability of treatment weighting.** The score used as a *weight*. A person who **did** receive the drug gets weight $(1/e)$; one who **did not** gets $(1/(1-e))$. This upweights the **surprising** people. If only one low-risk patient in ten ends up on celecoxib ($e = 0.1$), he is your only window into what the drug does to a low-risk body — so you clone him ten times (weight = $1/0.1 = 10$). The result is a **pseudo-population** in which the drug was handed out *independently of risk*: the risk→drug arrow erased by arithmetic, so a simple division now estimates the causal effect.

🧐 **ELI5 — IPTW** You wanted a fair coin-flip experiment; you got a rigged draw where sick people nearly always got the pill. IPTW doesn't throw anyone out — it *rebalances* by counting the rare, surprising people many times over, until each risk group is represented proportionately in both arms. You are rebuilding, on paper, the randomised trial nobody ran.

- **New-user design.** Follow people only from their *first* prescription, so you don't enrol survivors who already tolerated the drug for years and make it look safer than it is.
- **Negative controls.** Run the method on a drug-outcome pair with no plausible causal link; whatever non-null effect returns is the method's *residual bias*, measured.
- **Self-controlled designs.** Compare each person to themselves, on-drug vs off-drug; every fixed personal trait cancels automatically.

M1.3 The trap: they are all subtraction machines

Every method above has one job: **take a contaminated estimate and remove a known contaminant**. Feed one a dataset with zero contamination and it subtracts zero and returns the number it was given. The code runs, the model converges, the weights are sensible, the diagnostics are clean, the negative controls come back null — **and every one of those outputs is correct**. You have learned nothing about whether the method works.

On GiBleed, a correctly implemented propensity-score method and a catastrophically broken one produce the same answer. The dataset cannot tell them apart. The test has *zero* discriminating power — a metal detector at a doorway nobody carries metal through. "Perfectly calibrated" and "unplugged" are indistinguishable.

🧑‍🚒 **ELI5 — the lifeboat in the swimming pool** It floats! Everyone stayed dry! The test passed! But the thing the lifeboat was built to survive — the sea — was never introduced. GiBleed can validate that your code runs. It cannot validate that your method works.

M1.4 The second-order sting: the diagnostics are untested too

Overlap plots, weight distributions, positivity checks exist to catch the pathologies that break IPTW. With 68% uptake and no channelling, GiBleed has no extreme weights, no non-overlap, no positivity violations. So **the machinery for detecting when the method fails is untested too**. The alarm and the fire suppressant are both certified in a room with no fire.

M1.5 Why RR = 1.32 measures nothing

The 2×2 (355 / 1,489 / 124 / 726) gives $RR = 1.32$, $OR = 1.40$. The arithmetic is correct. But:

observed association = true causal effect + confounding + noise

Term 2 is zero by construction (no confounding generated). Term 3 is zero by construction (ordering 355/355, not one tie — §C4). So 1.32 is *entirely* term 1 — and term 1 was produced by a line in Synthea's configuration, not by a stomach.

🧑‍🚒 **ELI5 — the painted thermometer** You can read it precisely. It has a number on it. It reflects nothing outside itself. You are measuring a settings file, through the world's most elaborate instrument.

M1.6 The benchmarking consequence

Someone builds a new causal-inference method, validates it against the standard portable sandbox — GiBleed — and it runs, the estimate looks sensible, the negative controls behave. They write "*validated against GiBleed*." What they actually showed is that **their code executes end to end** — software testing, useful — reported as **method validation**, a claim about bias removal that was never tested because the bias was never there. The two look identical from outside: same green checkmarks. **Nobody is lying and nothing malfunctioned**. The failure is entirely in the inference from *passed* to *seaworthy*.

M1.7 Two caveats that make the claim more defensible

- **PS and IPTW only ever claimed to handle *measured* confounding**. Pre-empt the reviewer: GiBleed lacks even the measured kind, so it can't exercise the easy case the method was designed for. This strengthens the point.
- **"Teaches the mechanics perfectly" is a genuine compliment**. For learning *how to write* a CohortMethod package, GiBleed is ideal — fast, clean, and the absence of

noise makes code bugs visible. The failure is only in what people then *conclude*.

Paste-ready README block for M1

markdown

****Downstream damage – the most serious finding in this audit****

Every standard confounding-control method runs cleanly on GiBleed. Propensity scores, IPTW, new-user designs, negative controls, self-controlled designs – they execute, they converge, they return plausible numbers.

They have to. Each of these methods exists to *subtract a bias*. GiBleed contains none, so each one subtracts zero and returns the number it was handed. ****A method validated here has not been validated.**** A correct implementation and a broken one produce the same result, because there is nothing to detect.

Three consequences:

- ****RR = 1.32 is not a measurement.**** With no confounding and no noise, the risk ratio reflects the generator's parameter settings – not biology, not bias, not celecoxib.
- ****Benchmarking here is uninformative.**** A new causal-inference method tested against GiBleed will appear to work. That conclusion is unsupported: it shows the code executes, not that the method removes bias.
- ****The diagnostics are untested too.**** No positivity violations, no extreme weights, no overlap failures – the pathologies these checks exist to catch never occur.

A dataset built to teach a method for handling confounding contains no confounding. It teaches the mechanics perfectly and the reason for the mechanics not at all.

> It is a lifeboat tested in a swimming pool.

B — M2 · 100% osteoarthritis prevalence

2,694 of 2,694. Real-world symptomatic OA prevalence is roughly an order of magnitude lower. Every person has exactly one OA record.

M2.1 The finding

- **No unexposed comparator exists** — there is no OA-free group to compare against.
- **OA has zero variance**, therefore zero covariance with every other variable. It cannot confound, mediate, stratify, or be adjusted for. **Formally, it is not a variable.**
- Any model including OA as a covariate is unidentifiable in that dimension.
- Any comorbidity overlap involving OA is determined entirely by the other condition's

marginal. *Confirmed*: Otitis media $\cap$ OA = 2,025 = **all** 2,025 otitis patients.

M2.2 Why "nested" is the wrong word for it

A natural instinct is to describe this as otitis being *nested* within OA. Set-theoretically exact — but as an *explanation* it runs the arrow backwards, and it should be avoided.

OA is not 100% because otitis sits inside it. **Otitis sits inside it because OA is 100%**. The moment OA covers everyone, *every* condition is nested within it — trivially, all thirty of them. Nesting is not a fact about the pair; it's a fact about OA alone.

The word also carries technical baggage — nested case-control designs, nested random effects, and *informative* clinical nestings (all diabetic-retinopathy patients have diabetes, which encodes a real dependency). A reader who knows the term will hunt for the structure that justifies it, find none, and lose trust in the section.

The correct statement: OA is a constant — the set-theoretic form of which is that OA is the *sample space*. $A \subseteq \Omega$ for every A. That is why $|\text{otitis} \cap \text{OA}| = |\text{otitis}|$: not a discovery about comorbidity, an identity.

🧐 **ELI5** "Everyone in town came to the party" tells you nothing about *who* came. Ask "how many redheads?" and the answer is "all of them" — but you knew that the instant you heard *everyone*. The overlap isn't a measurement. It's the same sentence said twice.

M2.3 Two formalisms that survive a hostile reading

- **Information-theoretic.** $H(\text{OA}) = 0$, and mutual information is bounded above by the smaller entropy, so $I(\text{OA}; X) = 0$ for every X. The cross-tab is fully determined by the other marginal. There is no joint distribution — only a marginal, copied.
- **Linear-algebraic — softened to be exactly correct.** Coded 1 for everyone, the OA column is **identical to the intercept column**. The design matrix is not singular in some exotic way; it is **rank-deficient by one** because the same regressor was entered twice. Nothing blows up — R returns `(NA)` for the aliased coefficient; drop the intercept instead and OA simply *becomes* the intercept. The coefficient is not undefined — it is **not separable from the baseline**. Saying it this way pre-empts the reader who runs the model, sees it complete without error, and wonders what you meant.

M2.4 The 2,025 is a check, not evidence

That number is *arithmetically forced* by the 100% figure, so it cannot fail. Its value is as a **confirmation that passed** — it verifies the 100% figure is real and the person-level joins behave. Frame it as a check, or a careful reader notices it is circular.

Tightened block

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- ****No unexposed comparator exists.****
- ****OA is a constant, not a variable.**** Zero variance $\Rightarrow$ zero covariance with everything. It cannot confound, mediate, stratify, or be adjusted for.
- ****It cannot enter a model.**** Coded 1 for everyone, the OA column equals the intercept column. The design matrix is rank-deficient by one and the coefficient is not separable from the baseline.
- ****Every comorbidity overlap involving OA is forced.**** OA is the sample space so any condition is a subset by construction. *Verified:* otitis n OA = 2, = all otitis patients — and the same identity holds for all 30 conditions.

B — M3 · The GI-bleed differential is present, complete, and one-shot

Eight conditions occur **exactly once per person and never again**. Read as a clinician, seven of them name themselves — *this is the differential diagnosis of GI bleeding*, instantiated as a family of one-shot events.

	condition	people	prevalence	window
Upper GI	Peptic ulcer	802	29.8%	24–34
	Esophagitis	409	15.2%	31–44
	Angiodysplasia of stomach	388	14.4%	30–44
Lower GI	Diverticular disease	405	15.0%	32–47
	Polyp of colon	380	14.1%	31–46
	Ulcerative colitis	413	15.3%	31–45
Outcome	Gastrointestinal hemorrhage	479	17.8%	32–47
Indication	Osteoarthritis	2,694	100%	31–47

Downstream damage. *Nobody bleeds twice* — rebleeding is *the* clinical problem in GI hemorrhage and it does not exist here, so time-to-recurrence, rebleed risk, and secondary prevention are all unstudyable. *Etiology-specific risk is impossible* — each cause appears once, in isolation, no severity gradient, no progression. Ulcerative colitis with no flares; diverticular disease with no diverticulitis; peptic ulcer with no *H. pylori* eradication and no relapse.

⚠ **recs_per_person = 1.00** is not 100% prevalence. Different columns. Ulcerative colitis affects 15.3% of people — and all 413 of them got it exactly once. Only osteoarthritis is at 100% prevalence.

🧐 **ELI5 — why eight is a different finding from three** Three odd things is "*huh, that's odd.*" Eight odd things that are all the same kind of odd, in the same age bracket, that

together spell out a textbook page — that isn't odd. That's a machine leaving a **serial number**. You didn't find a bug. You found the shape of the tool that made it.

B — M4 · The age windows are clinically impossible

Every module member is diagnosed inside a ~15-year midlife window and **never after 47** — in a database that records other conditions out to **109**.

- **Angiodysplasia of stomach** is a disease of the *elderly* — aortic stenosis, chronic kidney disease, typically over 60. Here: 388 people aged **30-44**, never again.
- **Diverticular disease** rises steeply after 50, present in most of the population by 80. Here it **stops at 47**.
- **Polyp of colon** — screening begins at 45 *because that is when polyps appear*. Here they are **finished by 46**.
- **Osteoarthritis** — real incidence peaks around 55-75 and never turns off. Here: nothing new after **47**.

This is not censoring, and the proof is in the same table. Viral sinusitis is recorded from age **0 to 109**. The recording apparatus demonstrably works across the whole lifespan. When the module's diagnoses stop dead at 47 in that same database, it is a **true absence**.

Downstream damage. Age cannot confound anything, so age-adjusted models are unidentifiable within the module. Age-stratified analysis is meaningless — there are no strata above 47. Incidence-by-age curves are step functions with an off switch. Any study relying on age as an effect modifier finds no effect — **and the null is an artifact**.

B — M5 · Nobody dies

Zero deaths. 98 people over 100 (**3.6%**, against a real-world ~0.03%). The population reaches **110**.

Downstream damage. *Competing risk does not exist* — in real GI-bleed research death competes with rebleeding (patients who die cannot bleed again), so cumulative incidence and naive risk coincide here where in reality they never do. Survival analysis is vacuous — there are no events. *Immortal time bias is undetectable* because everyone genuinely is immortal. Any mortality-adjusted or death-censored design cannot be tested here.

 **ELI5 — the absorbing state that was never built** Death is the one room in the whole map that every human-population model has: a door in, no door out. GiBleed forgot to build it — so nobody in 2,694 people ever dies, including the 110-year-olds still catching sinusitis. (The modelling consequence is §C2.)

B — M6 · Prescribing does not resemble prescribing

68% celecoxib (real-world ~1-3%). **98.7% tetanus/diphtheria toxoid coverage**. One celecoxib exposure per person.

Downstream damage. Propensity-score distributions are unrealistic — no positivity violations, no extreme weights, no overlap failures, i.e. none of the pathologies PS diagnostics exist to catch (this is the §M1.4 sting, seen from the prescribing side). *Channelling is absent* — nobody was switched to a COX-2 selective *because* they were high-risk. **One exposure record per person compresses years of chronic NSAID therapy into a single event** — the exact mechanism by which GI risk accumulates, deleted. And 98.7% adherence makes adherence research and healthy-adherer bias unstudyable.

🧐 **ELI5** Celecoxib is a second-line, cardiovascular-risk-carrying, cost-constrained drug. Two-thirds of a general population being on it is like finding that two-thirds of a town drives the same rare, expensive car — before you look at anything else, you already know the town isn't real.

B — M7 • There is no comorbidity structure

Every pairwise overlap equals the smaller group **entirely**.

Downstream damage. Charlson and Elixhauser comorbidity indices are effectively constants. Multimorbidity research is impossible. Covariate adjustment is vacuous — there is nothing to adjust for.

🧐 **ELI5** A real 70-year-old with osteoarthritis has hypertension, or diabetes, or chronic kidney disease — usually more than one. Here the conditions don't *interlock*; each sits in its own sealed box. A body is a web of correlated failures. This is a filing cabinet.

C • The biomathematical lens

Treating each condition as a compartment in a deterministic system.

C — B1 • Eight absorbing states that cannot exist

C1.0 What a record actually is

One row in `(condition_occurrence)` means: *this person was observed in this disease state, and the system stamped it*. A record is a **footprint**. `(recs_per_person)` is "how many footprints did the average affected person leave in this room" — a question about *behaviour*. One footprint: stepped in, stepped out. Forty: they **lived** there.

C1.1 A disease is a place, not an event

A novice thinks of a diagnosis as an **event** (it happened, tick). A modeller thinks of a disease as a **state** — a place a body can occupy, linger in, leave, and return to. The record is the stamp collected each time you're seen there. Once disease is a place, the **arrows** between places are the object of study: a return arrow (recover then relapse), a **self-loop** (chronic, re-coded each visit). **The number of footprints is a direct readout of how many arrows got traversed.**



C1.2 Reading the graph off the count

Osteoarthritis — a knee wearing out for decades — is in any real record **a hallway a person paces for forty years**: first ache, first X-ray, follow-up films, injections, physio, the replacement, post-op checks. Dozens of footprints. GiBleed records it **once**. One footprint in a room someone lived in for forty years.

ELI5 — records-per-person as a fingerprint Count footprints. One footprint = someone stepped in once and stood there forever. Real arthritis is a person pacing a hallway for forty years — hundreds of footprints. **One footprint per person isn't a typo. It means the hallway isn't real.**

C1.3 The formal object, stated precisely

Model each history as a Markov process; a state's behaviour lives in its row of the **transition matrix**. A recurrent condition has a rich row (mass on recover, relapse, a self-loop) — those entries generate multiple records. The eight have a row with a single entry: the arrow *in*, nothing else.

Terminology caution. A *strict* Markov absorbing state has a self-loop of probability 1 — you stay forever, which in an EHR would generate *many* records. What GiBleed has is a **sink with no self-loop**: you enter and the record simply stops, as if the person vanished from the graph. Either way the signature is `(recs_per_person = 1.00)`, exactly. If you use "absorbing," add half a sentence that you mean it in the *record-generating* sense.

C1.4 Why one number proves nothing and the neighbours prove everything

Alone, `(1.00)` is defensible — a skeptic says "maybe these are just coded as single events." What kills that is that the **same query, same database, same moment** produced 23 other conditions that behave like life:

condition	recs/person
Viral sinusitis	6.43
Acute viral pharyngitis	3.92
Acute bronchitis	3.22
Otitis media	1.78
Sprain of ankle	1.41
Chronic sinusitis	1.02
the eight	1.00

Viral sinusitis at 6.43 is the **positive control**: caught, cleared, caught again, the record breathing exactly as recurrence should. The apparatus *can* represent repeats — it does so 23 times. So the eight welded to the floor cannot be blamed on the instrument. ($\overline{1.00}$ is also the hard **floor** — you cannot have fewer than one record per person who has the condition — so the eight are pressed flat against a boundary nothing else comes near.)

🧑 ELI5 — two rulers, one line The evidence isn't the number; it's the **distribution the number sits in**. One number proves nothing. You don't need a textbook — you need the neighbours.

C1.5 The four things that die, and exactly why

Each is a modelling family that needs **movement**, handed data with none.

1. **Multi-state / Markov models unidentifiable.** A rate is transitions ÷ time-at-risk. Zero transitions ⇒ zero numerator; the likelihood is **flat** along those parameters. *You cannot estimate the speed of a journey nobody took.*
2. **No sojourn times, no hazard of re-entry.** A sojourn time needs entry *and* exit timestamps (two footprints); one record gives one. The hazard of re-entry needs re-entries; there are none — and **rebleeding is the clinical problem**.
3. **Mixed models lose their random effects.** Splitting variance into between- vs within-person needs **repeated measures** — ≥ 2 observations per person. With exactly one, within-person variance has zero observations to be estimated from; it collapses into the residual and the random effect is unidentifiable.

🧑 ELI5 — one blood-pressure reading. To know how much your blood pressure bounces around *your own* average, someone has to measure it more than once. Measure it once and "your average" and "today's reading" are the same number. The bounce isn't small — it's **invisible**.

4. **Fan-out does not exist — and its absence is the artifact.** Joining a person to their condition records is normally **one-to-many**; naive $\text{COUNT}(*)$ counts *pairings*, and sicker patients have more rows, so the error **over-weights exactly the population**

you're studying. Every EHR analyst spends a career defending against this. Here the join is 1:1; fan-out never happens. **Real data fights you, and this data doesn't** — and frictionlessness is itself the fingerprint of a generator.

Through-line. `recs_per_person` is a measurement of the transition matrix. Life has a busy matrix and leaves many footprints; the eight have an empty matrix and leave exactly one. Every model above dies for the same reason — each estimates **flow**, and you cannot estimate the dynamics of a system frozen the instant it was created.

C — B2 • The death compartment was never built

Death is the **canonical absorbing state** of any human-population model. Every $S \rightarrow I \rightarrow R$ system, every Markov chain over a lifespan, every survival model has one. GiBleed's is missing entirely, while the population ages to 110.

Downstream limitations. This is an **open system with no exit** — population is not conserved in any meaningful sense. Any compartmental model with a mortality sink cannot be fitted. And the absence is **invisible to a conformance check: an empty table is a valid table**. (This is B1's "absorbing state" logic applied to the one state that *should* be absorbing and isn't even present — the mirror image of the eight states that are absorbing and shouldn't be.)

C — B3 • The hazard function has an off switch

Incidence is **zero before ~31, positive inside the window, and zero forever after** — in a system that demonstrably records events to 109.

C3.0 The hazard function from scratch

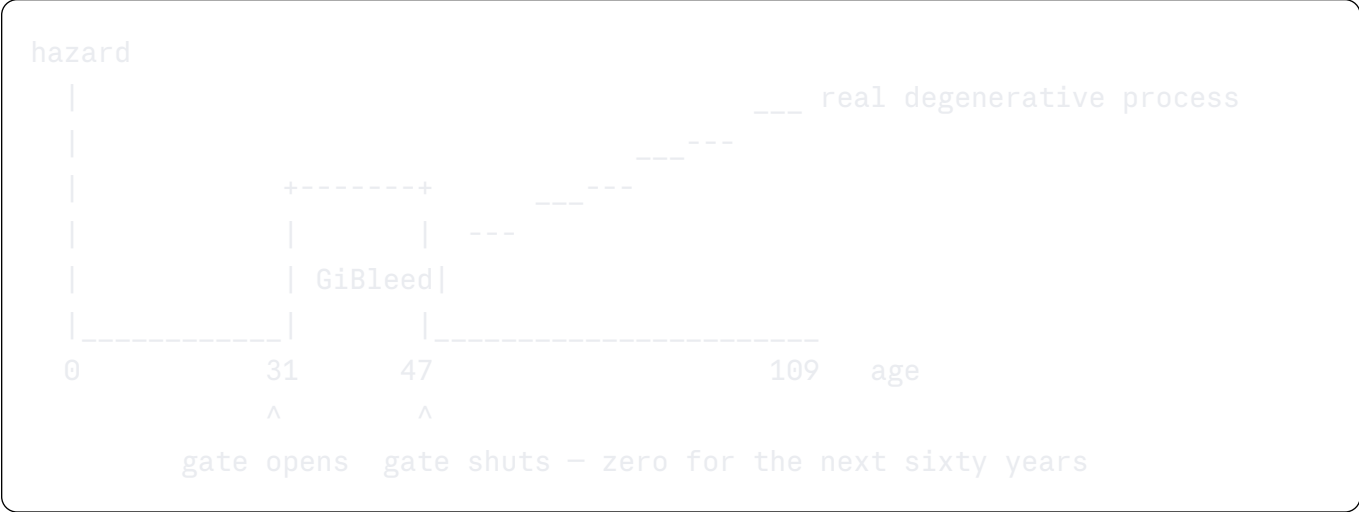
For anyone who doesn't yet have the disease: *given that you're still fine today, what's the chance you develop it in the next instant?* That conditional risk-right-now is the **hazard**. Two things define it: it is **conditional on still being at risk** (denominator = people who haven't got it yet), which makes it a property of the *process* not of the pool draining; and it is **instantaneous** — a rate at a moment.

 **ELI5 — speedometer, not odometer.** The odometer says how far you've come; the speedometer says how fast you're going *now*. Hazard is the speedometer of disease risk, and its *shape over a lifetime* is the fingerprint of the underlying physical process.

C3.1 Hazard shapes are signatures of mechanism

process	hazard shape
Wear-out / degeneration	rises with age and never stops
Random insult	roughly flat (memoryless)
Early failure	high at first, then falls

OA, atherosclerosis, most vascular disease are wear-out. **You cannot get from wear-out to a hazard of zero** — not "very low," *zero* — because that would say the damage stopped accumulating, that every knee became immune to further degeneration at 47 and stayed that way for sixty years while those same people kept walking into clinics with sinusitis.



C3.2 Why "zero after 47" is a claim, not an absence

The objection is **censoring** — maybe nobody was *watched* after 47. The audit closes it with the **positive control**: viral sinusitis recorded 0-109 in the same database. The window spans the whole lifespan. **It isn't that we stopped looking — it's that the process stopped happening.** That is what earns the phrase *true absence*.

C3.3 Why no parametric survival family can fit it

A parametric survival model chooses a **family** — a small set of allowed shapes with a couple of knobs:

family	shape	knobs
Exponential	constant	1
Weibull	a power of time — rises/falls/flat, but monotone and smooth	2
Gompertz	rises exponentially with age — the classic degeneration shape	2

None can produce zero → positive → zero. A **rectangular pulse** is outside the reachable space of all three.

🧠 **ELI5 — bendy rulers and cliffs.** You're fitting a ruler to a shape. Weibull and Gompertz bend one way, gently. GiBleed's hazard has two **vertical cliffs**. No adjustment of the knobs produces cliffs, because cliffs aren't in the family's vocabulary.

And the model still returns numbers — a fitted shape parameter with a standard error and a confidence interval, the best possible summary of a shape the family cannot represent. **Same silent failure as M1: the software succeeds, and the succeeding is the problem.** Mechanistically, real hazards *accumulate* (integrals of damage, no delete key); GiBleed's is

switched off by a gate — `if (age >= 30 && age <= 47)`. The rectangular pulse is what a conditional looks like plotted against age.

C3.4 The precision warning (this is correction #1)

An earlier draft claimed the age gradient was **inverted** — young people getting a disease of the old, backwards. **Wrong**, caught (by a question about the 110-year-old), corrected. Why requires the most confused distinction in epidemiology:

quantity	meaning	plumbing
Incidence	new cases per unit time	the tap
Prevalence	proportion who currently have it	the level in the tank

For a condition nobody recovers from and nobody dies of, **prevalence can never fall**. Cases only enter; nothing removes them. So even if incidence drops to zero, prevalence **plateaus**.



Prevalence goes 0% → k% and **stays** — monotone increasing, exactly as a chronic irreversible condition should. **Nothing about the prevalence curve is inverted.**

The pathology is two distinct defects in a *different* quantity:

- **Shifted young** — the tap opens at 31; real OA incidence peaks 55–75, angiodyspasia >60.
- **Truncated** — the tap shuts at 47 in a population living to 110; real degenerative incidence has no off switch.

Why the error was tempting: angiodyspasia in 30–44-year-olds *looks* flipped. But "flipped" would mean incidence *falling* as age rises. It doesn't — it's off, then on, then off. **The window was moved and clipped, not reversed.**

C — B4 · Determinism where stochasticity belongs

355 → 0. Exposure precedes outcome 355 times out of 355.

C4.1 The setup — there are three answers, not two

355 people both took celecoxib and bled. For each: *which came first?* Three possibilities —

drug first, bleed first, or **both on the same date** (neither, because a date without a timestamp can't order events within a day). Every case must land in one of two counting buckets, and **the comparison operator decides where same-day cases go**.

C4.2 What the `>=` does

The rule was effectively: `count as bleed_first` if the drug date is on **or after** the bleed date. Follow a same-day case: drug date = bleed date, so "on or after" is true, so it counts as `bleed_first`. **Ties fall into the `bleed_first` pile.**

```
drug ----o-----o---- bleed      -> drug_first    355 cases
      (drug then bleed)

      o both, same day              -> bleed_first     0 cases

bleed ---o-----o---- drug         -> bleed_first     0 cases
      (genuine reversal)           time ----->
```

C4.3 Why the zero is so damning

`bleed_first` is the **generous** bucket — it catches genuine reversals **and** every ambiguous same-day pairing. It caught nothing. So the single zero carries two claims: **zero reversals** (no one bled before their prescription) **and zero ties** (no one had both events on the same date — one tie would make the counter read 1).

🤖 **ELI5 — the biased referee.** A referee awards every close call — every genuine foul **and** every ambiguous one — to the away team. At full time the away team has zero. You don't review the tape. Under a rule that generous, **zero can only mean nothing happened at all.**

This is stronger than a bare "355/355": a skeptic asking *how were ambiguous cases handled?* has already been answered — they were handed to the other side, which still scored nothing. (Master-record correction #5: the tie count was not "unreported"; the `>=` had already proved it.)

C4.4 Why zero ties is the more shocking half

Most EHR records carry a **date, not a timestamp**, so same-day events are genuinely unorderable, and same-day collisions are *common* (patient presents bleeding, coder backfills drug and diagnosis into one encounter). With late coding, wrong coding, and copy-forward lists, real ordering runs **~95-98% with a scruffy tail**. GiBleed has **no tail**.

Perfect is the tell. 100% doesn't mean excellent data — it means nobody rolled dice. Real records are the residue of a messy physical process; that mess is the evidence that something actually happened in the world. Its absence is the signature of an arrow that was **drawn** rather than **observed**.

Analytic cost: same-day collisions are exactly what stress-test a cohort definition — index dates, lookback windows, tie-breaking. On this data that logic is never exercised.

C — B5 • Two independent signatures, one boundary, no overlap

`recs_per_person` and the age envelope were measured **separately**. They select **the same eight conditions**:

	the eight	the other twenty-three
records per person	exactly 1.00	1.02 - 6.43
age window	~15 years, none after 47	0 → 96-110

Nothing sits between them except **one** partially-scripted case: *E. coli* UTI (1.15 recs/person, ages 16-66).

- Two measurements agreeing this cleanly do not describe a disease. They describe a code path.
- 

ELI5 — two rulers, one line. You measured with a ruler: *how often does it recur?* You got a line — eight on one side, twenty-three on the other. Then a totally unrelated ruler: *at what ages?* Same line. Same eight. When two unrelated rulers draw the identical line, the line isn't in your measurement — **it's a seam in the thing you're measuring.**

D • Reverse-engineering the generator

No source code, module definitions, or documentation were read. Everything below was recovered from output alone — from 65,332 condition records and seven queries. Each recovered fact is paired with the evidence that forced it:

Recovered	Evidence
A discrete GI-bleed module of 8 conditions exists	two independent signatures select the same 8, no overlap
Its membership is the clinical differential — 3 upper, 3 lower, + the outcome	the identity of the eight
One-shot semantics — no recurrence loop	<code>recs_per_person = 1.00</code> , exactly, eight times
An onset age gate at ~30–32	min ages 30, 31, 31, 31, 31, 32, 32
A termination gate at ~44–47	max ages 44, 44, 45, 46, 47, 47, 47
Peptic ulcer runs a distinct younger gate (24–34)	~7 years below the rest of the module
Osteoarthritis is the eligibility scaffold , not a modelled disease	100% prevalence + identical topology + identical window
No mortality module wired into this scenario	<code>death</code> empty; ages to 110
Module boundaries are hard , not probabilistic	<i>E. coli</i> UTI at 1.15 is the only intermediate case in 31 conditions

Then it was checked. `cdm_source` independently confirms: **Synthea, CDM v5.3.1, ETL-Synthea, released 2019-05-25**. The design-intent reconstruction — inferred from clinical reasoning alone, *before* reading the metadata — was correct.

The dataset is a sentence.

```
100% osteoarthritis          -> the INDICATION (everyone NSAID-eligible)
68% celecoxib + aspirin + ibuprofen -> the EXPOSURES
the GI differential, all one-shot -> the ETIOLOGIES
gastrointestinal hemorrhage    -> the OUTCOME
the dataset is literally named GiBleed
```

Read the topology and you can read the design.

🧐 **ELI5 — the reframe that this makes possible** "Found errors in a dataset" is a *chore* — anyone can find bugs. "Reverse-engineered an undocumented module in a code generator, from its output, using clinical knowledge" is a *demonstration of skill*. Same work. The second framing only became available when the club hit eight and the members named themselves.

E • Why it matters in a federated ecosystem

This is where the argument stops being about GiBleed.

OHDSI's method is **federated**: an analytic package is written once and executed at many sites against local CDM instances. Patient-level data never moves; only aggregate results come back.

That architecture has a specific blind spot. If a site's ETL introduces a topology artifact — collapsing recurrent encounters into one record, gating an age range, dropping a mortality feed — the network **cannot see it**. The site's data passes conformance. The results come back well-formed. And cross-site heterogeneity gets attributed to *population differences* when it is actually *ETL differences*.

That is a confounder introduced by infrastructure, invisible to every tool currently pointed at the problem. The implication:

Data quality must be established at the source, not rescued during analysis.

A multidomain sanity check — structure *and* pathophysiology *and* topology — run at each node catches these before they enter the federation. Run afterwards, all you can do is discover that your sites disagree and guess why. The rescue is always partial, always late, and always leaves you arguing about whether the signal was ever real.

F • The metadata finding

`cdm_source` is **model-citizen metadata** — complete, honest, well-linked. It independently confirmed the design-intent reconstruction. It documents the data as "*synthetic, realistic (but not real)*."

And it is still not enough. The label says *realistic* — but never says **realistic for what**. No field states that OA was set to 100% as scaffolding, that `death` is empty, that GI hemorrhage is enriched ~180×, or that confounding was never generated.

The CDM answers "where did this come from?" It has no field that answers "what can I use it for?" Fitness-for-purpose has no home in the CDM.

 **ELI5 — the label problem.** A bottle labelled "*Liquid. Realistic (but not real) beverage*." Every word true, the label complete and honest. **And it still won't tell you whether you can drink it.**

Bonus deviation: `vocabulary_version = v5.0 18-JAN-19` — seven years old. Everyone learning OHDSI on Eunomia learns against a January 2019 vocabulary. (Also: Eunomia's SQLite stores dates as Unix epoch integers rather than the CDM-specified `DATE` type — a small conformance deviation, found by checking a column before trusting arithmetic on it.)

G • Prior work and the gap in the literature

The broad claim is not new, and it is not being made. Others established that synthetic health data lacks clinical realism, at far larger scale:

Source	Finding
Chen et al. (2019), <i>BMC Med Inform Decis Mak</i>	Validated Synthea vs real-world quality measures in a 1.2M Massachusetts cohort; limited ability to model heterogeneous post-intervention outcomes. doi:10.1186/s12911-019-0793-0
Wagner & Blacketer (2024), OHDSI Symposium	1.16M Synthea California vs real-world via Achilles; limited modelling diversity hinders fidelity, especially for chronic disease.
"Synthetic but Not Realistic" (2026), arXiv:2606.08903	Synthetic-data evaluation is dominated by statistical-similarity measures that don't reflect clinical validity.
Kahn et al. (2016), <i>eGEMs</i>	The conformance/completeness/plausibility framework GiBleed passes.
MIRACUM (2024)	Fitness-for-purpose assessment is an active field.

What is new here is narrower — four defensible claims:

1. **GiBleed specifically.** Prior evaluations used million-person generated populations; this audits the **2,694-person default teaching database** where most practitioners' mental model of "what OMOP data looks like" is formed.
2. **Disease-state topology as a data-quality prior** — records-per-person as a signature of transition structure, derived from compartmental-modelling formalism.
3. **Module-boundary recovery from output alone** — an undocumented generator module identified via two independent clinical signatures, without source access.
4. **Design-intent reconstruction as a validation method** — recovered from clinical reasoning, then tested against the documented source. Because Synthea's design is documented, this is falsifiable. **It held.**

The gap is the field's own admission, not an assertion. A 2024 fitness-for-purpose study found data-quality experts predominantly hold computer-science backgrounds, *"potentially lacking clinical or statistical expertise."* The ML-DQA framework concedes determinations often rest on a single clinical expert, and that conformance/completeness/plausibility *"may not capture all aspects of data quality."*

 **ELI5 — why admitting you were scooped makes you more credible** Two people describe a fish. One: *"biggest ever."* The other: *"pretty big — though I first said twice that, then I measured it."* You believe the second one. About everything. Forever. The literature check made the artifact **smaller and better.**

H • Errors made and corrected

The most important section for credibility. Anyone can audit someone else's work; applying the same standard to your own, in public, unprompted, is the entire thing being claimed.

Technical corrections

#	Error	Correction
1	"Inverted age gradient"	Wrong. Prevalence goes 0%→ <i>k</i> % and <i>stays</i> — monotone increasing, never inverted. Correct: shifted young and truncated . Caught by the question about the 110-year-old.
2	Called peptic ulcer 802/802 "the outcome side"	Wrong. GI hemorrhage (192671) is 479/479. Peptic ulcer is a separate module member.
3	Tagged the ordering query "reconstructed"	It was verbatim in Cell 23 all along.
4	Criticised <code>LIKE '%hemorrhage%'</code> as "not a cohort definition"	My code, not yours. You used <code>condition_concept_id = 192671</code> from the start. Retracted.
5	Claimed the same-day tie count was "unreported"	The <code>>=</code> folds ties into <code>bleed_first</code> ; <code>0</code> proves zero reversals AND zero ties — stronger than claimed.
6	Filed <code>AND</code> under modifiers	It's an operator — two booleans in, one out.
7	"Swap <code>LIMIT 15</code> for <code>HAVING >= 300</code> "	Ambiguous. <code>LIMIT</code> was the culprit; <code>HAVING</code> would have removed nothing (smallest row was 510 people).
8	Roadmap said "cells 9–17"	It's 9–15 ; 16 is the closer.
9	Charts said three at 1.00	It's eight . Regenerated.
10	ORCID <code>0000-0003-4159-3853</code>	Cannot verify independently. ⚠️ UNRESOLVED.

Framing corrections (v1 → v2)

Was	Became
Assumed novelty	Literature check found the broad claim was made first — replaced with four narrow claims.
Read findings as defects	<i>"None of these are bugs. They are scaffolding."</i>
Tone as critique	Tone is curiosity — a newcomer critiquing OHDSI's teaching dataset reads as an attack regardless of correctness.
Risk of implying commissioned work	Label " self-directed " everywhere.
Over-weighting the artifact	It supports your case; it isn't your case. Must not crowd out peer-reviewed work.

I • Cross-cutting methodology

The transferable core — these recur across every finding above.

1. **The positive control converts absence into evidence.** Viral sinusitis at 0–109 does this twice: it proves the apparatus spans the lifespan (so B3's silence after 47 is real, not censoring) *and* that it can represent recurrence (so B1's flat 1.00 is real, not a coding convention). Without a positive control, every "we found nothing" is indistinguishable from "we didn't look."
2. **The contrast carries the argument, never the single number.** One number proves nothing — you need the neighbours. (The generalisation of the positive control.)
3. **The dangerous failures are the silent ones.** *Unusable* is honest — you try, it fails, you stop. **Dangerous** means your propensity scores run, your negative controls behave, your Weibull returns a shape parameter with a CI, and you conclude your method works. Every finding here lands in the second category.
4. **Perfection is diagnostic of absence, not quality.** 355/355 with no ties. $\text{Var}(X) = 0$. `recs_per_person` = exactly 1.00. Fan-out that never happens. Real data fights you; the fighting is the evidence.
5. **State precisely and the claim survives; state loosely and it dies to a plot.** "Inverted" → refutable in ten seconds; "shifted young and truncated" → correct and mechanistic. "Nested" → borrowed connotations; "OA is the sample space" → explains four bullets at once. "Singular design matrix" → overstates; "rank-deficient by one, not separable from the baseline" → exactly what R does.
6. **Pre-empt the strongest objection inside the text.** PS/IPTW handle only *measured* confounding — say so before a reviewer does, and note GiBleed lacks even that. Same move as flagging the 2,025 as a check rather than evidence.

J • Reference figures

All from the master record; do not re-derive from memory.

quantity	value
Persons	2,694
Condition records (total)	65,332
People with ≥ 1 condition	2,694 (no healthy controls)
Deaths	0
People over 100	98 (3.6%) · real-world ~0.03%
Max recorded age	110
The module (all recs/person = 1.00):	
Osteoarthritis — <i>indication</i>	2,694 (100%), ages 31–47
Peptic ulcer — upper GI	802 (29.8%), ages 24–34
Esophagitis — upper GI	409 (15.2%), ages 31–44
Angiodysplasia of stomach — upper GI	388 (14.4%), ages 30–44
Diverticular disease — lower GI	405 (15.0%), ages 32–47
Polyp of colon — lower GI	380 (14.1%), ages 31–46
Ulcerative colitis — lower GI	413 (15.3%), ages 31–45
Gastrointestinal hemorrhage — <i>outcome</i>	479 (17.8%), ages 32–47
Positive control: Viral sinusitis	6.43 recs/person, ages 0–109
Only intermediate case: <i>E. coli</i> UTI	1.15 recs/person, ages 16–66
Celecoxib users	1,844 (68%) · real-world ~1–3%
Td toxoid coverage	2,660 (98.7%)
GI bleeders on celecoxib	355 / 479 (74.1%)
Bleeders with no celecoxib exposure	124 (25.9%)
Celecoxib users who bled	355 / 1,844 = 19.3%
2×2 cells (a/b/c/d)	355 / 1,489 / 124 / 726

quantity	value
RR / OR	1.32 / 1.40
Ordering test	drug_first 355 · bleed_first (incl. same-day) 0
Otitis media $\cap$ OA	2,025 = all otitis patients
Generator	Synthea · ETL-Synthea · CDM v5.3.1 · released 2019-05-25
Vocabulary	v5.0 18-JAN-19 (~7 years old)
Method	seven SQL queries, ~90 seconds, no ML, no privileged access

K • The core lines

The mathematics tells me what shape a disease should leave in data. The medicine tells me when the shape is wrong.

None of these are bugs. They are scaffolding.

None of them violate a range check.

The CDM answers "*where did this come from?*" It has no field for "*what can I use it for?*"

One number proves nothing. You don't need a textbook — you need the neighbours.

Not "I found bugs." Anyone can find bugs. I reverse-engineered an undocumented module in a code generator, from its output, using clinical knowledge — then checked the literature, found others had made the broader point first, and said so out loud.